

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q23-2334

Paragould Reciprocating 138 kV  
December 29, 2024

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2334. The Interconnection Customer proposes a 136.8 MW solar photovoltaic facility interconnecting at the Paragould Reciprocating 138 kV substation in the Virginia area. The request was received on July 16, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 510 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,326,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility           | Contingency                   | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------|-------------------------------|-------------|--------------|----------|
| Shockoe – Moonstone 138.0 kV | Big Hanaford – Plaza 138.0 kV | 110.4       | 237          | 21.5     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------|---------------------------------------------|---------------|-----------------------|
| Shockoe – Moonstone 138.0 kV | Rebuild line with higher-capacity conductor | \$9,280,000   | 36% (\$3,326,000)     |
|                              | Total allocated to project                  |               | \$3,326,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.